Phase diagram of networks grown with triadic closure rules

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We study a network that grows by four independent processes: nodes arrive at rate α_+ and attach to a random node, random nodes are deleted at rate $\alpha_- < \alpha_+$, open triads close at rate β_+, and closed triads open at rate β_-. At $\beta_+ = \beta_-$ the triangles become a critical queue, and for $\beta_- \geq \beta_+$ the network is a tree whose degrees stay bounded, because every closure that is reversed strips a link from the centre of the triad. For $\beta_+ > \beta_-$ the outcome depends on how the open triad is found. As order parameter we take the fraction $P_\triangle$ of all triangles that lie in the largest cluster of triangles sharing links. If triads are drawn uniformly from the whole network, the triangles condense: $P_\triangle \to 1$ while the cluster spans a vanishing fraction of the nodes and holds every net closure link. Without opening the condensate is a near-clique of order $\sqrt{n}$ nodes, which appears beyond an onset size that grows roughly as u^{-5}, $u = \beta_+/(\alpha_+ + \beta_+)$. Node deletion does not remove it: the condensate is a steady state rebuilt from young nodes, not a relic of the oldest ones. Opening disperses the triangles into finite clusters beyond a turnover $\beta_-/\beta_+ \approx 0.75$, and deletion moves this boundary towards $\beta_- = \beta_+$. Local versions of the closure — a node introducing two of its friends, or befriending a friend of a friend — never condense; their triangles form one cluster spanning a finite fraction of the nodes, and the friend-of-a-friend rule reproduces the local-search model $LS(n, 1)$. The two kinds of closure organize the network differently. Local closure produces assortative, hierarchical community structure, with a number of communities that grows in proportion to n. Census closure produces a strongly assortative split between a club and a periphery that barely link to each other. The tree phase is weakly disassortative and has no community structure.

I. INTRODUCTION

Triadic closure, a link forming between two nodes that share a neighbour, is the mechanism behind the high clustering of social networks [1–6]. In an evolving university e-mail network, pairs two steps apart were about 30 times more likely to form a tie than pairs three steps apart, and the probability of a new tie grew with the number of common contacts [7], as it does under the census rule (w) below. In the local-search model $LS(n, \ell)$ [4, 13, 14] with $\ell = 1$ a newcomer picks a random node j, links to it and to a random neighbour of j. Every arrival closes one triangle, so arrival and closure are the same event, and no link is ever lost.

In a social system these are separate processes. People join and leave at their own rates, acquaintances are introduced to each other at another, and ties lapse at yet another. We therefore let four processes run side by side: arrival at rate α_+, deletion of a node at rate α_-, closure of an open triad at rate β_+, and opening of a closed triad at rate β_-. Opening is the reverse of closure, the transition from a link back to a potential link that Ref. [14], Ch. 3, sets aside. Jin, Girvan and Newman [16] combined meetings biased towards pairs with mutual friends with the decay of unrenewed ties, and found clustered networks with strong community structure. Bianconi *et al.* [17] showed that growth by triadic closure alone generates communities, and that heterogeneous node fitness replaces them with a few superhubs that attract most of the links. Their model, which they name triadic closure, is one specific way of closing triads: closure is tied to arrival, a newcomer linking to a random node and, with probability p, to its neighbours, as in the local-search model [4, 6]. On a fixed population, link creation through friends of friends balanced by random link decay gives a discontinuous transition between a sparse and a dense society, with coexistence and hysteresis [8], and a birth rate that grows with the number of common neighbours gives bistability [9]. Closure through a uniformly random wedge, our rule (w) below, was analysed on a fixed vertex set to bound the growth of degrees [10]. In the random triadic process each triad may close only if its triple is present in a random hypergraph of density p, and the graph becomes complete above the sharp threshold $p = 1/(2\sqrt{n})$ [11]. Static triadic closure on a fixed backbone is exactly solvable for its motif densities [12]. Here nodes keep arriving, the rates are uniform, and the question is what the balance between arrival, closure and opening alone produces.

Growth is what separates this model from the closure-and-decay models of Refs. [8, 9], and it changes the outcome in two ways. First, in a fixed population every node is statistically equivalent, so the network as a whole is either sparse or dense, and the two states compete as alternatives: bistability, coexistence, hysteresis. In a growing network nodes differ by age, and the dense state appears locally, on a vanishing fraction of nodes that coexists with a sparse majority. Bistability between global states becomes condensation. The condensate is not a memory of the first nodes: it survives the deletion of nodes, which eventually removes every one of them, and is rebuilt from younger ones (Sec. VI B). Second, with per-pair rates and a fixed population the dense state is a finite fraction of

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all pairs, whereas arrivals bring a reservoir of new nodes and a density that can be held fixed. We show that this requires total rates in fixed ratio: per-triad rates, the analogue of the per-pair rates of Ref. [9], have no stationary density once nodes arrive (Sec. III).

Section II defines the model and four ways of choosing the triad. Section III fixes the density and finds the first transition, at $\beta_+ = \beta_-$. Section IV gives the uncorrelated mean field. Section V introduces the triangle-cluster order parameter and lays out the phase diagram, and Secs. VI–VIII characterize its three phases. Section IX treats the other closure rules, and Sec. X the communities.

II. THE MODEL

The network starts from a path of three nodes. With $Z = \alpha_+ + \alpha_- + \beta_+ + \beta_-$, each event is

- an *arrival*, with probability α_+/Z: a new node links to a uniformly random node;
- a *deletion*, with probability α_-/Z: a uniformly random node is removed together with all its links;
- a *closure*, with probability $u_+ = \beta_+/Z$: an open triad a–c–b (a, b unlinked) gets the link (a, b);
- an *opening*, with probability $u_- = \beta_-/Z$: a closed triad a–c–b (a, b linked) loses the link (a, b).

Since Z only sets the clock, we measure rates in units of α_+ and keep $\alpha_- < \alpha_+$, so that the network grows. For $\alpha_- > \alpha_+$ the number of nodes fluctuates around a small value and there is no large-n limit. How a triad is chosen defines the rule. Each closure rule has a mirror-image opening rule:

(w) *random triad*: an open (closed) triad drawn uniformly from the whole network. The centre of an open triad is then chosen with weight $\propto k_c(k_c - 1)$. A link belongs to as many closed triads as it has triangles, t_e, so opening removes a link with probability $\propto t_e$. The closure step is that of the wedge picking model [10];

(p) *random potential link*: closure picks a uniformly random unlinked pair with a common neighbour (the rule simulated in Ref. [14], Ch. 3); opening picks a uniformly random link with $t_e \geq 1$;

(c) *introducer*: a random node c with $k_c \geq 2$ picks two of its neighbours and links them if they are unlinked (closure), or unlinks them if they are linked (opening);

(f) *friend of a friend*: a random node a walks $a \to j \to l$, $l \neq a$, and links to l if unlinked (closure), or unlinks from it if linked (opening).

Rules (w) and (p) need a census of the network; (c) and (f) use only a node and its neighbourhood. If the selected configuration does not qualify, the selection is repeated. A closure that finds no open triad becomes an arrival, and an opening that finds no closed triad does nothing. Deletion is skipped when fewer than four nodes remain. Opening a triad never disconnects the network, since a and b remain joined through c; deletion can. Rule (w) is the main object of this paper. Rules (p), (c) and (f) are treated, for comparison, in Sec. IX. The reference is $LS(n, 1)$, which adds one node and two links per step.

The simulation keeps adjacency lists, a hash set of links, a Fenwick tree over the triad weights $k_c(k_c - 1)/2$, and, when opening is on, the triangle count t_e of every link in a second Fenwick tree. Triangle counts are updated incrementally and were checked against full recounts, with and without deletion. A run stops when the network first holds n nodes.

III. DENSITY AND THE FIRST TRANSITION

Write $r_\pm = \beta_\pm/\alpha_+$ for the number of closures and openings per arrival, and

$$\Delta = r_+ - r_- = \frac{\beta_+ - \beta_-}{\alpha_+} \tag{1}$$

for the net growth, the number of links that closure adds, net of opening, per arrival. Each arrival adds a node and a link, each closure adds a link, and each successful opening removes one. A deletion removes a node and, on average, $\langle k\rangle$ links. If every opening finds a closed triad, the links grow at rate $\alpha_+ + \beta_+ - \beta_- - \alpha_- \langle k\rangle$ and the nodes at rate $\alpha_+ - \alpha_-$, and a stationary density $E/n = \langle k\rangle/2$ requires

$$\langle k\rangle = \frac{2(1 + \Delta)}{1 + \alpha_-/\alpha_+}. \tag{2}$$

To keep the density constant the rates must be held in fixed ratio. Without opening or deletion, $\beta_+ = \alpha_+$ reproduces the density of $LS(n, 1)$, $\langle k\rangle = 4$. All runs below agree with Eq. (2); for example, at $\Delta = 1$ and $n = 10^6$, $\langle k\rangle = 2.65$–2.69 at $\alpha_- = \alpha_+/2$ and 2.10–2.15 at $\alpha_- = 0.9\,\alpha_+$, against 2.67 and 2.11.

Total rates are not the only reading of the rates. If instead every node spawns a newcomer at rate α_+ and every open triad closes at rate β_+ ($\beta_- = \alpha_- = 0$), the closure rate is $\beta_+ W$, with W the number of open triads, and W/n is not a constant of the dynamics. In the uncorrelated mean field the density is $\langle k\rangle = 2/(1 - x)$, with $x = \beta_+ m/\alpha_+$ and $m = \langle k(k - 1)\rangle/\langle k\rangle$, which is finite only for $x < 1/2$. The self-consistent solution exists only up to $\beta_+/\alpha_+ = 0.067$, where $\langle k\rangle = 2.58$. Simulations run away for $\beta_+/\alpha_+ \geq 0.06$ within 10^3–5×10^4 nodes. At $\beta_+/\alpha_+ = 0.04$, $\langle k\rangle$ holds at 2.23 until the network condenses near $n = 10^6$ (Sec. VI); then $W \sim n^{3/2}$, closures outpace arrivals, and $\langle k\rangle$ reaches 2.67 at 1.6×10^6

and keeps rising. Since condensation appears to happen at every $u > 0$, per-triad rates should have no stationary density at any $\beta_+ > 0$, and total rates in fixed ratio are the consistent choice. On a fixed population, per-pair birth rates that grow with the number of common neighbours give two stable states, a sparse and a dense one [9]. With arrivals the dense state takes over.

In the rest of this section $\alpha_- = 0$.

Equation (2) cannot hold for $\beta_- \geq \beta_+$, since it would give $\langle k\rangle \leq 2$ while the network stays connected, $E \geq n-1$. The openings must then fail often enough to keep $E \geq n-1$. Close to a tree each closure creates one triangle and each opening destroys one. The triangle count is therefore a queue with arrival rate β_+ and service rate β_-. It grows linearly in n for $\beta_+ > \beta_-$, is critical at $\beta_+ = \beta_-$, and stays $O(1)$ for $\beta_- > \beta_+$. In the last case a fraction $(\beta_- - \beta_+)/\beta_-$ of the openings find nothing to remove.

The simulations follow this exactly. At $\beta_+ = \alpha_+$ and $n = 10^6$ the fraction of failed openings is 0.093, 0.202, 0.501, 0.751 and 0.900 for $\beta_-/\beta_+ = 1.1$, 1.25, 2, 4, 10, against the predicted 0.091, 0.200, 0.500, 0.750, 0.900. At $\beta_+ = \beta_-$ the network holds $T = 317$ triangles at $n = 10^6$, of order $\sqrt{n}$ as for a critical queue. The excess degree $\langle k\rangle - 2 = 2\Delta$ vanishes linearly at $\beta_- = \beta_+$, and the triangles per node fall from 0.75 at $\beta_-/\beta_+ = 0.5$ to 3×10^{-4} at 1 and 10^{-5} at 1.1.

The parameter space is therefore cut by the plane $\beta_- = \beta_+$. Below it the network is sparse with extensive triangles and density given by Eq. (2); above it the network is a tree. Deletion removes triangles but never creates them, so the queue argument, and with it this boundary, holds at any α_-; the tree then breaks into a forest. Since only ratios of rates matter, we use three coordinates: the net growth Δ, which with α_-/α_+ fixes the density, the turnover β_-/β_+, and the deletion α_-/α_+.

IV. UNCORRELATED MEAN FIELD

Here $\alpha_- = 0$. Let a closure pick e of its two endpoints with probability proportional to degree and the other $2-e$ uniformly. In the absence of correlations, the degree of a node grows with the number of arrivals as

$$n\frac{dk_i}{dn} = 1 + r_+(2-e) + \frac{r_+ e}{\langle k\rangle}k_i - r_-\frac{2t_i}{3T/n}, \qquad (3)$$

where the last term is opening under rule (w): node i is an endpoint of the removed link with probability $2t_i/3T$, with t_i its number of triangles. Rules (w), (c) and (p) reach both endpoints through a link, so $e = 2$. Rule (f) starts from a uniformly random node, so $e = 1$.

Without opening, $k_i \sim (n/n_i)^b$ with $b = e\,u/2$, $u = \beta_+/(\alpha_+ + \beta_+)$, and

$$\gamma = 1 + \frac{1}{b} = 1 + \frac{2}{e\,u}. \qquad (4)$$

The density cancels from the preference because of Eq. (2). At $u = 1/2$, rules (w), (c) and (p) give $\gamma = 3$, and rule (f) gives $\gamma = 5$, which is the exponent of $LS(n,1)$ itself [14]. In $LS(n,1)$ the newcomer is the uniform endpoint and the neighbour of j the preferential one. Separating arrival from closure leaves the exponent unchanged as long as one endpoint of the closure stays uniform.

With opening under rule (w), and clustering falling as $1/k$, we have $t_i \propto k_i$ and $\sum_i t_i = 3T$, so the opening term is $-2r_-k_i/\langle k\rangle$. Then

$$n\frac{dk_i}{dn} = 1 + \Big(1 - \frac{2}{\langle k\rangle}\Big)k_i, \qquad (5)$$

and $\gamma = 1 + \langle k\rangle/(\langle k\rangle - 2)$, independent of β_- at fixed density. In this approximation turnover does nothing. All the structure below therefore comes from correlations.

V. THE PHASE DIAGRAM

Condensation is a statement about where the triangles go, and the order parameter should say so without a choice of scale. Call two triangles adjacent when they share a link, and a *triangle cluster* a connected set of adjacent triangles: a 3-clique community in the language of clique percolation [19, 20]. Every closure creates at least one triangle, so the clusters record where the closures went. Let T be the number of triangles in the network and $T_{\max}$ the number of triangles in its largest triangle cluster. We take as order parameter the fraction of all triangles in the largest cluster,

$$P_\triangle = \frac{T_{\max}}{T}, \qquad (6)$$

the analogue for triangles of the fraction P of nodes in the giant component of a percolating network. We use with it the fractions $P_\bullet$ and $P_{\bullet-\bullet}$ of all nodes and all links that the largest cluster spans. If the triangles are scattered over finite clusters, $P_\triangle \to 0$. If $P_\triangle$ stays finite the triangles percolate, and $P_\bullet$ separates extended percolation, $P_\bullet$ finite, from condensation, $P_\bullet \to 0$. In $LS(n,1)$, for instance, a single cluster holds every triangle and every node, $P_\triangle = P_\bullet = 1$. The clusters are found by union-find over links, joining the three links of every triangle.

We ran rule (w) at $\Delta = 1/2$ and 1, at $\alpha_-/\alpha_+ = 0$ and 1/2, and at $\beta_-/\beta_+ = 0$, 0.5, 0.67, 0.75, 0.8, 0.85, 0.9, 0.95, with $n = 10^4$, 10^5, 10^6. We then refined the turnover near the boundaries in steps of 0.02, with four seeds at 10^5 and two at 10^6, scanned $\alpha_-/\alpha_+ = 0.1$–0.9 at $\Delta = 1$, and followed one point to 10^7. The result is Fig. 1. At small turnover $P_\triangle$ rises to one with n; at large turnover it falls to zero, below 10^{-3} at 10^6. Between the two, the curves for different n cross. We call a point condensed when $P_\triangle$ rises between 10^5 and 10^6 (or exceeds 0.9) and dispersed when it falls. Along every line of fixed Δ and α_-/α_+ the label changes once, which brackets the boundary $(\beta_-/\beta_+)_c$:

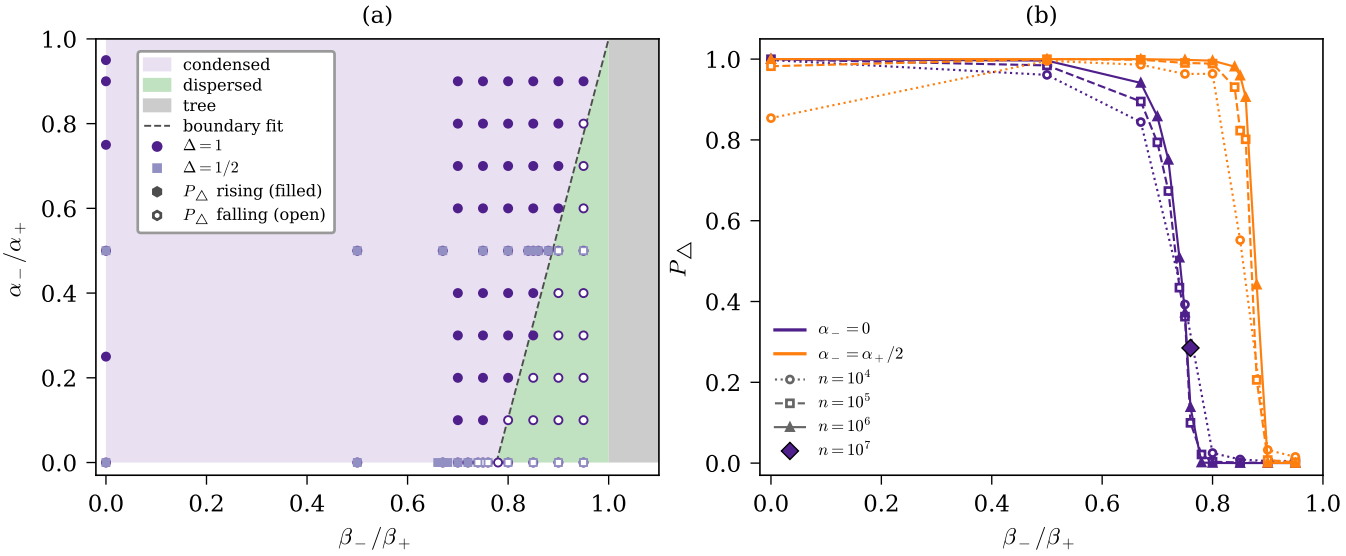


FIG. 1. Phase diagram of rule (w) from the triangle-cluster order parameter $P_\triangle$. (a) The plane of turnover β_-/β_+ and deletion α_-/α_+, at net growth $\Delta = 1$ (circles) and $\Delta = 1/2$ (squares). A symbol is filled when $P_\triangle$ rises between $n = 10^5$ and 10^6, or exceeds 0.9 at 10^6, and open when it falls. Shades: condensed (purple), dispersed (green) and tree (grey, $\beta_- \geq \beta_+$) phases. Dashed line: the fit of Eq. (7) to the $\Delta = 1$ boundary. (b) $P_\triangle$ against β_-/β_+ at $\Delta = 1$, without deletion (purple) and with $\alpha_- = \alpha_+/2$ (orange), for $n = 10^4$ (dotted lines, open circles), 10^5 (dashed lines, open squares) and 10^6 (solid lines, filled triangles), means over one to six seeds. Diamond: a network of 10^7 nodes at $\beta_-/\beta_+ = 0.76$, $\alpha_- = 0$.

α_-/α_+	$\Delta = 1$	$\Delta = 1/2$
0	0.76–0.78	0.72–0.74
0.1	0.75–0.80	
0.2	0.80–0.85	
0.3, 0.4	0.85–0.90	
0.5	0.88–0.90	0.88–0.90
0.6, 0.7, 0.8	0.90–0.95	

At $\alpha_-/\alpha_+ = 0.9$ every turnover up to 0.95 is condensed. The point closest to the boundary, $\Delta = 1$, $\alpha_- = 0$, $\beta_-/\beta_+ = 0.76$, has $P_\triangle = 0.10$, 0.14, 0.29 at $n = 10^5$, 10^6, 10^7. It condenses, but slowly, as expected near a critical point. Its four seeds at 10^5 give $P_\triangle = 0.05$–0.19, spread out but not bimodal.

Three features stand out. First, the boundary depends only weakly on density: without deletion it lies at 0.73 for $\Delta = 1/2$ and 0.77 for $\Delta = 1$, and at $\alpha_- = \alpha_+/2$ the two coincide within the resolution of 0.02. Second, deletion moves the boundary towards $\beta_- = \beta_+$. At $\Delta = 1$ the brackets follow

$$1 - \left(\frac{\beta_-}{\beta_+}\right)_c \simeq 0.22\left(1 - \frac{\alpha_-}{\alpha_+}\right), \tag{7}$$

a least-squares fit to the midpoints. It falls inside every bracket but two, which it misses by 0.006 ($\alpha_-/\alpha_+ = 0.3$) and 0.005 (0.8). If Eq. (7) holds up to $\alpha_- \to \alpha_+$, the dispersed phase closes there. Third, near the boundary $P_\triangle$ varies smoothly with turnover and drifts slowly with n, with no jump at the sizes simulated.

There are three phases:

- *condensed*, $\beta_- < (\beta_-)_c$: $P_\triangle \to 1$ and $P_\bullet \to 0$. One triangle cluster holds essentially all the triangles and every net closure link, on a vanishing fraction of the nodes (Sec. VI);
- *dispersed*, $(\beta_-)_c < \beta_- < \beta_+$: extensive triangles and $\langle k \rangle$ from Eq. (2), but $P_\triangle \to 0$ (Sec. VII);
- *tree*, $\beta_- \geq \beta_+$: $\langle k \rangle = 2$, $O(1)$ triangles, and bounded degrees (Sec. VIII).

The degree distributions of the three phases are in Fig. 2b.

VI. THE CONDENSED PHASE

A. Without opening: a dense core

With $\beta_- = \alpha_- = 0$ the only parameter is $u = \beta_+/(\alpha_+ + \beta_+)$, and $\langle k \rangle = 2/(1-u)$. Under rule (w) at $u = 1/2$, $n = 10^6$, the $10^3 = \sqrt{n}$ highest-degree nodes are linked to one another with density 0.81 and hold 20% of all links. The 100 richest form a complete graph, and all of them were born among the first few thousand arrivals. This share does not shrink with size: $\phi(\sqrt{n}) = 0.176$, 0.192, 0.202 at $n = 10^4$, 10^5, 10^6, while the density climbs to 0.71, 0.77, 0.81. The same holds for every $u \geq 0.3$. The core is a dense block of $\Theta(\sqrt{n})$ nodes inside a sparse network. It is the seed of the giant triangle cluster, which holds $P_\triangle = 0.99996$ of the triangles and $P_{\bullet-\bullet} = 0.501$ of the links at $n = 10^6$, that is,

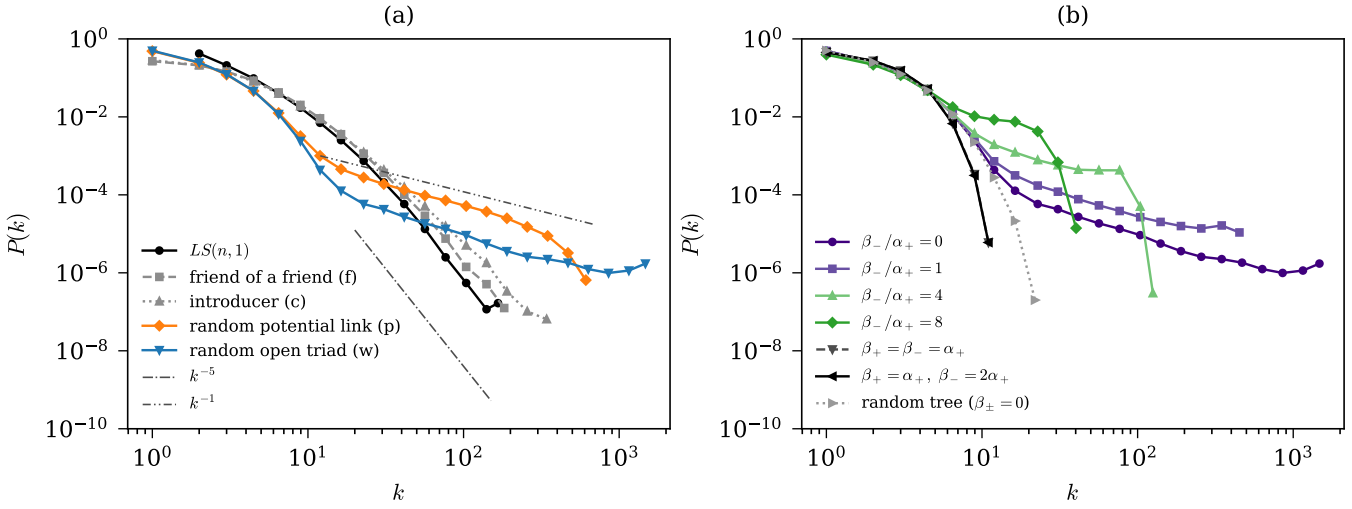


FIG. 2. Degree distributions in logarithmic bins, $n = 10^6$. (a) The closure rules without opening at $\langle k\rangle = 4$ ($u = 1/2$). Reference lines, k^{-5} and k^{-1}. (b) Rule (w) across the phase diagram: at $\langle k\rangle = 4$ with increasing turnover, $\beta_-/\alpha_+ = 0$, 1 ($\beta_-/\beta_+ = 0$, 0.5; condensed, purple) and 4, 8 ($\beta_-/\beta_+ = 0.8$, 0.89; dispersed, light and dark green); at the tree transition $\beta_+ = \beta_- = \alpha_+$ (dashed); in the tree phase $\beta_+ = \alpha_+$, $\beta_- = 2\alpha_+$ (black); and the uniform random tree, $\beta_\pm = 0$ (dotted).

every closure link. The cluster spans $P_\bullet = 0.074$, 0.027, 0.0096 of the nodes at $n = 10^4$, 10^5, 10^6, so $P_\bullet \sim n^{-0.45}$ (Fig. 4a): about $10\sqrt{n}$ nodes, the core and the peripheral nodes tied to it by a closure. The number of triangles therefore grows as $T \sim n^{3/2}$. The same scales appear in the random triadic process, which below its threshold stops at $O(n^{3/2})$ links and above it fills the whole graph [11]; here the census rule fills only a $\sqrt{n}$ block, and the transitivity stays near 0.5 while the mean local clustering vanishes (Table I). The degree distribution has a second maximum at $k \sim \sqrt{n}$ (Fig. 2a), and the degree of the oldest nodes saturates with age.

The hub degree follows from the core's own triads. Once triads centred in the core dominate the census, a core node a is an endpoint of a closure with probability w_a/W. Here $w_a \simeq \rho\sqrt{n}\,K$ counts the open triads through its $\rho\sqrt{n}$ core neighbours, and $W \simeq \sqrt{n}\,K^2/2$. Hence $n\,dK/dn \simeq 2\rho r_+ n/K$ and

$$K^2 \simeq c\,r_+\,n, \tag{8}$$

TABLE I. Rules without opening at $u = 1/2$ ($\langle k\rangle = 4$), $n = 10^6$. b: age exponent, fitted over $10^2 < n_i < 10^5$; $b_{\rm MF} = eu/2$; $\mathcal{T}$: transitivity; $\bar{C}$: mean local clustering; r_k: degree assortativity [21]; $\phi(\sqrt{n})$: club fraction of the $\sqrt{n}$ highest-degree nodes; ρ: link density among them.

rule	b	$b_{\rm MF}$	$k_{\max}$	$\mathcal{T}$	$\bar{C}$	r_k	$\phi(\sqrt{n})$ (ρ)
$LS(n,1)$	0.27	1/4	169	0.25	0.69	0.05	0.001 (0.002)
(f)	0.31	1/4	203	0.24	0.37	0.28	0.002 (0.008)
(c)	0.37	1/2	391	0.22	0.42	0.16	0.002 (0.007)
(p)	0.66	1/2	684	0.03	0.02	0.52	0.026 (0.11)
(w)	0.89	1/2	1619	0.48	0.008	0.65	0.20 (0.81)

with c of order 4ρ. The data give $k_{\max}^2/(r_+ n) = 2.8$, 2.7, 2.6, 2.5 at $u = 0.3$–0.6, $n = 10^6$, drifting down by about 10% per decade.

Onset. At small u a long normal plateau precedes the condensate. Along it T/n is constant and $k_{\max}/\sqrt{n}$ falls. The plateau ends abruptly: T/n turns up, and $k_{\max}/\sqrt{n}$ rises back to its condensed value (Fig. 3a). Taking the onset size n^* as the first n with $T/n \geq 1$, we find

u	0.08	0.10	0.12	0.15	0.20	0.25
$\ln n^*$	15.4	14.5	13.3	12.4	11.1	9.4

with seed-to-seed spread 0.1–0.5. Over $0.1 \leq u \leq 0.3$ a power law, $n^* \propto u^{-5.4}$, fits with rms 0.17 in $\ln n^*$. An essential singularity, $\ln n^* = 6.1 + 0.87/u$, fits with rms 0.41. Fitted before the $u = 0.08$ runs existed, these two forms predicted $\ln n^* = 15.7$ and 17.0 for $u = 0.08$. Two runs to 3×10^7 both condensed at 15.4, and five runs stopped at 3×10^6 had not condensed yet, as both forms require. At $u = 0.05$ nothing happens up to 10^7; the power law puts the onset at 8×10^7. Plotted against $1/u$, the onset curve bends down (Fig. 3b). A finite threshold $u_c > 0$, where n^* would diverge, would bend it up. The data thus favour condensation at every $u > 0$, with an onset size that becomes astronomical as $u \to 0$. A threshold below $u = 0.08$ cannot be excluded.

Mechanism. Under rule (w) a node is an endpoint of a closure with probability w_a/W, where $w_a = \sum_{c\in\partial a}(k_c - 1) - 2t_a$ counts the open triads ending at a. This rate is set by the degrees of a's neighbours, not by its own. In the uncorrelated limit $w_a \propto k_a$, which gives Eq. (3). But a hub of degree K is the centre of $\sim K^2/2$ triads, and each closure centred on it raises the degree of two of its neighbours. A two-variable caricature captures the feedback. Let s be the summed degree of the hub's neigh-

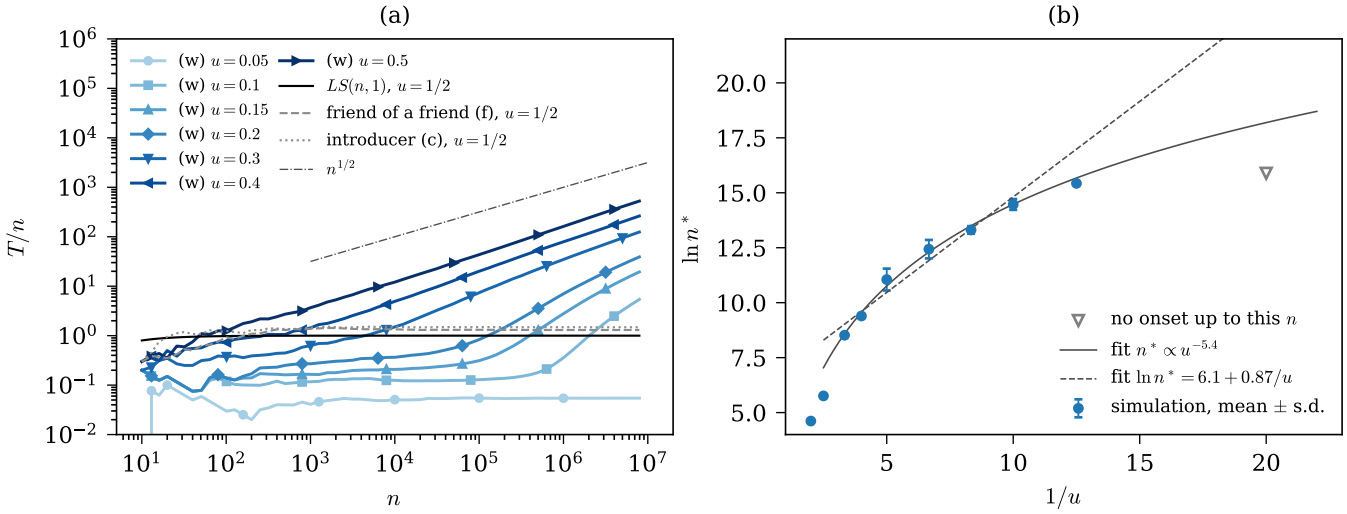


FIG. 3. Condensation without opening. (a) Triangles per node T/n along single trajectories to $n = 10^7$: rule (w) at $u = 0.05$–0.5 (blue lines with symbols, light to dark), and the local rules and $LS(n, 1)$ at $u = 1/2$ (grey and black). Dash-dotted line, $n^{1/2}$. (b) Onset size n^*, the first n with $T/n \geq 1$, mean and standard deviation over 5–6 seeds for $0.1 \leq u \leq 0.25$ and one or two trajectories otherwise. Lines are fits over $0.1 \leq u \leq 0.3$: a power law (solid) and an essential singularity (dashed). The point at $1/u = 12.5$ ($u = 0.08$, two runs to 3×10^7) was not used in the fits. Triangle: $u = 0.05$, no onset up to $n = 10^7$.

bours, $x = \ln n$, and $\epsilon = r_+/\bar{w}$, with $\bar{w}$ the mean number of open triads per node. Then $K' \simeq \epsilon s$ and $s' \gtrsim \epsilon K^2$, so $K'' \simeq \epsilon^2 K^2$. This blows up at a finite $x^* \sim \epsilon^{-1} K_0^{-1/2}$ for any $\epsilon > 0$, so the caricature has no threshold either. The blow-up stops only when the core dominates W, which gives Eq. (8). A symptom of the feedback is visible before the onset: in the normal plateau the five largest hubs keep a constant local clustering of ≈ 0.1 for every u between 0.02 and 0.1, instead of the $C \sim 1/k$ hierarchy of the local rules.

B. With deletion: a steady state

In a growing network the hubs are the oldest nodes, so the condensate might be a relic of the first arrivals rather than a state of the dynamics. Deletion separates the two. At rate $\alpha_- > 0$ every node is eventually removed: the network holds $n \propto (\alpha_+ - \alpha_-)t$ nodes, each deleted at rate α_-/n, so a node that arrived when a fraction s of all arrivals so far had taken place survives with probability $s^{\alpha_-/(\alpha_+ - \alpha_-)}$. Among the surviving nodes the median of s is $2^{-(1-\alpha_-/\alpha_+)}$: 0.71 at $\alpha_- = \alpha_+/2$ and 0.97 at $\alpha_- = 0.95\,\alpha_+$.

We ran rule (w) without opening, at $u = \beta_+/(\alpha_+ + \beta_+) = 0.3$ and $1/2$ and $\alpha_-/\alpha_+ = 0$, 0.25, 0.5, 0.75, 0.9, 0.95, up to $n = 10^6$. The condensate survives at every deletion rate. At $u = 1/2$ and $n = 10^6$, $P_\triangle \geq 0.9998$ for all α_-, the cluster holds the closure links, $P_{\bullet-\bullet} = 0.47$–$0.50$, and it spans $P_\bullet = 0.0023$, 0.0013, 0.0010 of the nodes at $\alpha_-/\alpha_+ = 0.5$, 0.9, 0.95. Its share falls with size as without deletion, $P_\bullet \sim n^{-0.45}$ at $\alpha_- = \alpha_+/2$ (Fig. 4a), and the largest degree still grows as $\sqrt{n}$: $k_{\max} = 71$, 268, 986 at $n = 10^4$, 10^5, 10^6 for $\alpha_- = 0.95\,\alpha_+$, against 1619 at 10^6 without deletion. At $u = 0.3$ deletion delays the onset, from below 10^4 to about 10^5–10^6 at $\alpha_-/\alpha_+ \geq 0.9$, but does not prevent it.

The condensate is young. The median arrival fraction s of the $\sqrt{n}$ highest-degree nodes is 0.001 without deletion, and 0.34, 0.70, 0.90, 0.96 at $\alpha_-/\alpha_+ = 0.5$, 0.75, 0.9, 0.95, against 0.71, 0.84, 0.93, 0.97 for a random surviving node. At strong deletion the hubs are as young as the population. The condensate is therefore a steady state: core nodes are deleted, and census closure promotes others from the periphery to replace them.

C. With opening: a sparse cluster

Opening hits a link in proportion to its triangles, so it falls mainly on the core. The core thins out. At $\langle k \rangle = 4$ ($\Delta = 1$) and $n = 10^6$, the share of links among the $\sqrt{n}$ richest nodes falls from 0.20 to 0.09, 0.02, 0.005 at $\beta_-/\beta_+ = 0$, 0.33, 0.5, 0.67. The triangle cluster does not dissolve, however. At $\beta_-/\beta_+ = 0.5$, $P_\triangle = 0.96$, 0.98, 0.996 at $n = 10^4$, 10^5, 10^6, and the cluster holds 99% of the net closure links, whose share of all links is $\Delta/(1+\Delta)$ (Fig. 4b). It spreads over more nodes, but their share still vanishes: $P_\bullet \sim n^{-\lambda}$ with $\lambda = 0.44$, 0.35, 0.25 at $\beta_-/\beta_+ = 0$, 0.5, 0.67, and $\lambda \approx 0.2$ at 0.74–0.75 (Fig. 4a). Turnover thus turns the dense core into a sparse cluster that stays localized. The largest degree follows. Between $n = 10^6$ and 8×10^6, $k_{\max} \sim n^\theta$ with $\theta = 0.52$, 0.44, 0.33 at $\beta_-/\beta_+ = 0.33$, 0.5, 0.67, against $\theta = 1/2$ at $\beta_- = 0$. The degree distribution loses its second maximum, and its tail ends at progressively smaller k (Fig. 2b, purple).

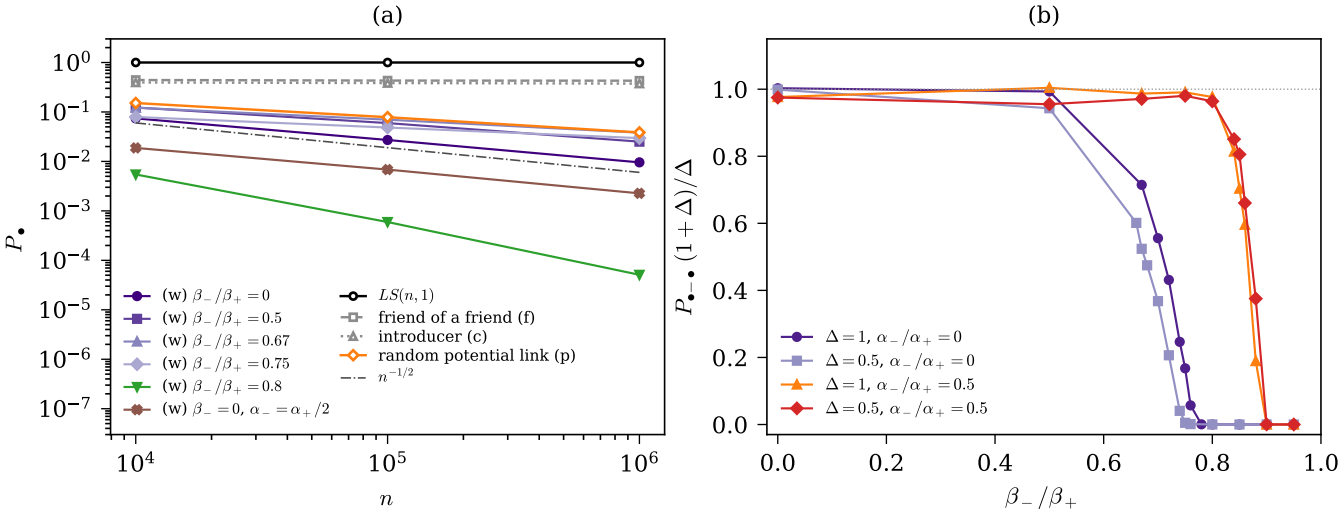


FIG. 4. The largest triangle cluster. (a) Fraction $P_\bullet$ of the nodes that it spans, against n: rule (w) at $\Delta = 1$ without deletion for $\beta_-/\beta_+ = 0$–0.75 (condensed, purple, dark to light) and 0.8 (dispersed, green); rule (w) at $\beta_- = 0$, $\Delta = 1$, $\alpha_- = \alpha_+/2$ (brown crosses); and, without opening at $u = 1/2$, rule (p) (orange diamonds), the local rules (grey) and $LS(n, 1)$ (black). Dash-dotted line, $n^{-1/2}$. Means over two seeds at $n = 10^4$ and one or two at larger n. (b) Fraction of the net closure links inside the cluster, $P_{\bullet-\bullet}(1+\Delta)/\Delta$, against β_-/β_+ at $n = 10^6$, for $\Delta = 1$ and 1/2 with $\alpha_-/\alpha_+ = 0$ and 0.5. Dotted line: all net closure links.

Approaching the boundary, the cluster first loses closure links and then triangles. Without deletion the share of net closure links inside it is 0.71, 0.43, 0.17 at $\beta_-/\beta_+ = 0.67$, 0.72, 0.75 for $\Delta = 1$, while $P_\triangle$ is still 0.94, 0.75, 0.37. With $\alpha_- = \alpha_+/2$ the cluster keeps 96–100% of them up to $\beta_-/\beta_+ = 0.8$ for both densities.

VII. THE DISPERSED PHASE

Above the boundary the triangles break into finite clusters. At $\Delta = 1$, $\alpha_- = 0$ and $n = 10^6$ there are 0.14, 0.10, 0.08 triangles per node at $\beta_-/\beta_+ = 0.8$, 0.85, 0.9, decreasing only slowly with size (0.22, 0.18, 0.14 at $\beta_-/\beta_+ = 0.8$ and $n = 10^4$, 10^5, 10^6). The largest cluster holds $P_\triangle < 10^{-3}$ of them at 10^6. At $\beta_-/\beta_+ = 0.8$ it has about 50 nodes at every size, so $P_\bullet \sim 1/n$ (Fig. 4a, green). The closure links are spread over the whole network.

Hubs remain near the boundary but grow slowly. For $\Delta = 1$ and $n = 10^5$, 10^6, $k_{\max} = 74$, 130 at $\beta_-/\beta_+ = 0.8$ ($k_{\max} \sim n^{0.23}$ up to 8×10^6), 49, 73 at 0.85, and 35, 41 at 0.9. The degree distribution falls off steeply beyond $k \approx 40$ (Fig. 2b, green), and the assortativity decreases with turnover, from 0.51 to 0.30 between $\beta_-/\beta_+ = 0.89$ and 0.94.

VIII. THE TREE PHASE

For $\beta_- \geq \beta_+$ nearly every triangle is opened soon after it forms. One time in three the opening removes the link just added, which undoes the closure. The other two times it removes one of the two links at the centre c, whose link passes to one of c's neighbours, so c loses one link. Under rule (w) the centre is drawn with weight $k_c(k_c-1)$. A node therefore loses links at a rate quadratic in its degree and gains them, as a neighbour of a centre, at a rate linear in it. The degrees are confined.

The data confirm this (Fig. 2b). At $\beta_+ = \alpha_+$, the largest degree is 11–13 for every $\beta_- \geq \beta_+$, at both $n = 10^5$ and 10^6. The uniform random tree of the same size reaches 24. The distribution bends down on the log-log plot. The ratio $P(k)/P(k-1)$ falls from 0.6 at $k = 2$ to 0.2 at $k = 10$, where the uniform random tree has a constant 1/2. The assortativity is -0.14, against -0.001 for the uniform random tree: leaves attach preferentially to the few nodes of larger degree that the pruning of triad centres leaves behind. The critical network at $\beta_+ = \beta_-$ has the same degree distribution as the tree phase. The two differ only in their $O(\sqrt{n})$ versus $O(1)$ triangles.

IX. OTHER CLOSURE RULES

Potential-link rule (p). Rule (p) also condenses, although without a dense core. Its largest triangle cluster holds $P_\triangle = 0.97$, 0.98, 0.99 of the triangles at $n = 10^4$, 10^5, 10^6 ($u = 1/2$) and $P_{\bullet-\bullet} = 0.51$ of the links, every closure link, on a share of the nodes that falls as $P_\bullet \sim n^{-0.29}$ (0.15, 0.078, 0.039; Fig. 4a). The links concentrate on the oldest nodes (Fig. 2a). The 1% richest nodes hold 6.5%, 18% and 37% of all links at $n = 10^4$, 10^5, 10^6 ($u = 1/2$). Its core is sparse, with density 0.1 among the $\sqrt{n}$ richest nodes and transitivity 0.03–0.017.

Its largest degree grows as about $n^{0.35}$. Without the m-fold weight of pairs sharing m neighbours there is no drive towards a clique, but the links still concentrate. The mean-field exponent of Ref. [14] for this rule ($\gamma = 2.62$ at $u = 1/2$) assumes a closure rate per potential link. That makes the total closure rate proportional to the number of potential links, which is not the rule simulated, and which does not conserve the density (Sec. III).

Local rules (c) and (f). The local rules never condense. Their triangles percolate on an extended cluster: at $u = 1/2$ the largest one holds $P_\triangle = 0.83$ (f) and 0.77 (c) of the triangles and spans $P_\bullet = 0.43$ and 0.37 of the nodes at $n = 10^6$, nearly the same as at 10^4 (0.44, 0.40; Fig. 4a). In $LS(n, 1)$ every new triangle contains the existing link between j and its neighbour, which already lies in a triangle, so a single cluster holds everything, $P_\triangle = P_\bullet = 1$. Without opening (Fig. 2a and Table I), they follow Eq. (4) up to corrections from correlations and a five-decade window. The age exponent b is 0.31 for (f) (mean field 1/4) and 0.37 for (c) (mean field 1/2), against 0.27 for $LS(n, 1)$ (exact 1/4). Their mean local clustering stays near 0.4.

With opening at $\langle k\rangle = 4$, rule (f) keeps its age exponent at 0.31–0.34 for $\beta_-/\alpha_+ = 0$ to 4, so it stays in the $LS(n, 1)$ class. Its mean clustering falls from 0.37 to 0.12. Rule (c) develops a broader tail ($k_{\max} = 391$ to 1287, age exponent 0.37 to 0.46), but the 1% richest nodes still hold only 2% of the links. On approaching $\beta_- = \beta_+$ at $\beta_+ = \alpha_+$ the hubs of rule (f) fade ($k_{\max} = 95$, 37, 17 at $\beta_-/\beta_+ = 0.5$, 0.8, 0.9), and at $\beta_+ = \beta_-$ it reaches the same critical tree ($k_{\max} = 12$, $T = 147$). For $\beta_- > \beta_+$ we did not run the local rules. Their opening starts from a random node and must find one of $O(1)$ triangles, which takes too long to simulate.

X. COMMUNITIES

Triadic closure is known to generate communities [16, 17]. We look for them by statistical inference with graph-tool [22]. We fit two models: the planted-partition model [23], which describes assortative communities only and returns one group when there are none, and the nested degree-corrected stochastic block model (SBM) [24, 25], which describes any mixing pattern. Both select the number of groups by minimum description length, so, unlike modularity maximization [26], they do not find groups in random graphs. As a check, each network is rewired by the configuration model and refitted. Every rewired network gives a single group, so the structure below does not come from the degree sequence. Table II gives the results at $n = 10^5$ and 10^4.

The two kinds of closure give opposite pictures. The local rules and $LS(n, 1)$ have strong community structure. They split into about 10^3 assortative groups at $n = 10^5$ with modularity $Q = 0.90$–0.95, and the number of groups grows nearly in proportion to n, by factors of 7 to 9 between 10^4 and 10^5. The SBM nests them in 7 to 9 levels, a hierarchy of communities, as in the triadic-closure models of Refs. [16, 17].

Rule (w) has no such hierarchy. The SBM finds two to six groups at every turnover up to $\beta_-/\beta_+ = 0.89$, and at most three levels. One group holds 73–95% of the nodes. The others contain the rich nodes: their mean degree is 9 to 39, against $\langle k\rangle = 4$. With two groups, 93–95% of the links lie inside a group, so the club and the periphery barely link to each other. The modularity of this split, 0.4, measures a club against a periphery, not a set of communities. At $\beta_- \le \beta_+/2$ the planted-partition fit also cuts the periphery into 25–35 groups, but only half the links fall inside them ($Q \approx 0.3$). Rule (p), whose core is sparse, gives the same club-and-periphery picture. Turnover shrinks the periphery's share, from 95% of the nodes at $\beta_- = 0$ to 60% at $\beta_-/\beta_+ = 0.89$. The split is therefore not a signature of the condensate. It survives into the dispersed phase, where the closure links are no longer held by one triangle cluster but still join nodes of high degree: the assortativity is still 0.51 at $\beta_-/\beta_+ = 0.89$ (Sec. VII). Further in, at $\beta_-/\beta_+ = 0.94$, there is no structure. The critical network, the tree phase and the uniform random tree also have none: a tree with bounded degrees offers no evidence for groups.

Bianconi *et al.* [17] found that triadic closure makes communities, and that heterogeneous fitness destroys them in favour of superhubs that share the rest of the nodes. Here the same replacement happens with uni-

TABLE II. Communities at $n = 10^5$ (in parentheses, $n = 10^4$). $B_{\rm pp}$: groups of the planted-partition model [23]; Q: modularity of that partition; S_1: fraction of nodes in its largest group; $\bar{k}_s$: mean degree of the nodes outside the largest group; $B_{\rm sbm}$ and L: groups at the bottom level and number of nontrivial levels of the nested degree-corrected SBM [25]. Rule (w) at $\langle k\rangle = 4$; $\beta_-/\beta_+ \le 0.67$ lie in the condensed phase and $\beta_-/\beta_+ \ge 0.8$ in the dispersed phase. Every configuration-model rewiring of these networks gives $B_{\rm pp} = B_{\rm sbm} = 1$.

network	$B_{\rm pp}$	Q	S_1	$\bar{k}_s$	$B_{\rm sbm}$	L
$LS(n, 1)$	1870 (216)	0.95	0.002	4.0	2435	9
(f), $u = 1/2$	1265 (154)	0.90	0.005	4.0	1863	7
(c), $u = 1/2$	1049 (148)	0.92	0.005	4.0	2128	9
(p), $u = 1/2$	3 (8)	0.43	0.94	36	3	1
(w), $\beta_-/\beta_+ = 0$	25 (2)	0.27	0.95	39	4	2
(w), $\beta_-/\beta_+ = 0.33$	26 (3)	0.28	0.94	32	4	2
(w), $\beta_-/\beta_+ = 0.5$	35 (10)	0.31	0.87	18	3	1
(w), $\beta_-/\beta_+ = 0.67$	2 (7)	0.43	0.92	29	6	3
(w), $\beta_-/\beta_+ = 0.8$	2 (2)	0.42	0.81	13	2	1
(w), $\beta_-/\beta_+ = 0.86$	3 (2)	0.41	0.73	9.4	2	1
(w), $\beta_-/\beta_+ = 0.89$	2 (2)	0.37	0.60	7.0	2	1
(w), $\beta_-/\beta_+ = 0.94$	1 (1)	0	1	—	1	0
(w), $\beta_+ = \beta_-$	1 (1)	0	1	—	1	0
(w), $\beta_- = 2\beta_+$	1 (1)	0	1	—	1	0
random tree	1 (1)	0	1	—	1	0

form rates, when closure draws its triads from the whole network instead of from the neighbourhood of a node.

XI. CONCLUSIONS

Once the clock is fixed the model has three parameters: the net growth $\Delta = (\beta_+ - \beta_-)/\alpha_+$, which with the deletion α_-/α_+ sets the density, and the turnover β_-/β_+. The fraction $P_\triangle$ of triangles in the largest triangle cluster is the order parameter. It is built from the objects that the closure rule creates, and it needs no choice of scale. The phase diagram has two boundaries.

- At $\beta_- = \beta_+$ the triangles become a critical queue. Below it they are extensive and $\langle k \rangle$ follows Eq. (2). Above it the network is a tree whose degrees stay bounded, because a reversed closure strips a link from the centre of the triad, and the centre is chosen with weight k^2.
- At $(\beta_-/\beta_+)_c \approx 0.73$–$0.77$ without deletion the triangles stop percolating. Below it, one triangle cluster holds essentially all the triangles and every net closure link on a vanishing fraction of the nodes. The cluster is a dense near-clique of order $\sqrt{n}$ nodes without opening, and opening spreads it over more nodes, $P_\bullet \sim n^{-\lambda}$ with λ falling from 0.45 towards 0.2. Above it the triangles are spread over finite clusters. Deletion moves the boundary towards $\beta_- = \beta_+$, approximately as Eq. (7).

The condensate is a steady state, not a relic of the first nodes. It survives deletion rates up to $\alpha_- = 0.95\,\alpha_+$, at which its members are as young as the population.

Condensation requires a census rule. Both census rules condense: rule (w) builds a dense core, and rule (p) a sparse one. When the triad is found locally, a random node introducing two friends or befriending a friend of a friend, the triangles percolate on a cluster that spans a finite fraction of the nodes, and the network stays scale-free and clustered with $\gamma = 1 + 2/(eu)$ at $\beta_- = 0$. The friend-of-a-friend version at $\beta_+ = \alpha_+$ is the faithful generalization of $LS(n, 1)$: same density, same exponent $\gamma = 5$. Local closure also builds a hierarchy of assortative communities. Census closure replaces it with a club and a periphery that barely link to each other, and an assortativity of 0.5–0.65 that persists into the dispersed phase.

The condensate differs from Bose–Einstein condensation in fitness models [15] and from superlinear preferential attachment [18]. There a single node captures a finite fraction of the links. Here a growing set of nodes does, every one of them with a degree that diverges with n. Without opening the set is a dense block of $\Theta(\sqrt{n})$ nodes, a sparse graph with a dense spike in graphon language.

Open questions. (i) A controlled theory of the onset $n^*(u)$, power law versus essential singularity, and of its seed-to-seed distribution. (ii) The nature of the condensation boundary. The $P_\triangle$ curves for $n = 10^4$–10^7 cross smoothly and the seeds are not bimodal, which points to a continuous transition, but its exponents, and the form of $(\beta_-/\beta_+)_c$ as a function of Δ and α_-/α_+, are open. Closure balanced by decay gives discontinuous transitions with hysteresis on fixed populations [8, 9]; quenches in β_- would test for hysteresis here. (iii) A theory of the localization exponent λ of the cluster. (iv) The degree distribution of the tree phase, which the k^2 erosion argument should give in closed form. (v) The phase diagram of rule (p) with opening. (vi) Mixtures of local and census closure, which should move both the onset and the boundary continuously.

Reproducibility. All results come from `tc.c` in the `triadic_closure` folder of the code repository (modes `l, w, p, c, f` and `r` for per-triad rates; the optional sixth and seventh arguments are u_- and α_-/Z), with triangle clusters from `trclust.c`. The scans are `scan.py`, `runs_ts.sh`, `runs_onset.sh` and `onset.py` (no opening); `scan_rm.py`, `runs_rm_ts.sh`, `cross.py` and `phase.py` (with opening); `opgrid.py`, `bnd.py`, `delgrid.py` and `rulesclust.py` (triangle clusters and deletion); and `comm.py` (communities, run with graph-tool). `mkfigs.py` draws the figures. Summaries are in `data/`.

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